

Netflix Data Analysis Using POWERBI

By

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Problem Statement:

Analyze Netflix's content and customer viewing behavior to derive meaningful business insights. The goal is to clean and visualize the data to understand customer engagement, content popularity, and subscription trends.

Domain Introduction

The **OTT (Over-The-Top) streaming industry** has transformed how users consume entertainment content globally. Platforms like Netflix leverage data analytics to understand viewer behavior and optimize content delivery. Analyzing user engagement, content trends, and subscriptions helps improve customer satisfaction and business growth.



Project Introductions

This project focuses on analyzing Netflix content and customer viewing behavior using Excel and Power BI.

The aim is to generate actionable insights on engagement, content popularity, and subscription patterns.

Objective of the Project

The objective is to clean, transform, and analyze Netflix data to identify patterns in user behavior and content performance. It also aims to support data-driven decision-making for content strategy and customer engagement



Data Cleaning & Transformation (Excel + Power BI)

Data Consistency Check

Handling Missing Values

Identified missing values in **Director**, **Cast**, and **Country** columns.

Replaced missing values with "**Unknown**" or "**Not Available**".

Ensured no blank fields remain for accurate analysis.

Standardizing Date Format

- Standardized **date_added** column into a consistent format (DD-MM-YYYY).
- Converted text dates into proper **Excel date format**.
- Ensured uniformity for time-based analysis.

Categorizing Rating Values

- Cleaned and standardized **rating values** (e.g., TV-MA, PG, R).
- Grouped ratings into categories like **Kids**, **Teens**, **Adults**.
- Ensured consistent classification for analysis.

Customer Demographics Analysis

- Grouped customers into age ranges: **13-18, 19-30, 31-45, 46-60, 60+**.
- Counted number of customers in each age group.
- Compared viewing activity to identify **most active age segment**.

Insights

- Identified **dominant age group** with highest number of viewers.
- Found that **young adults (19-30)** often contribute most to viewership.
- Older age groups show **lower engagement comparatively**.

Analysis

- Helps Netflix target **content and marketing by age group**.
- Supports creation of **age-specific recommendations**.
- Improves **customer engagement strategies**.



Subscription Type vs Watch Time

- Grouped data by **subscription type (Basic, Standard, Premium)**.
- Calculated **total/average watch time** for each subscription tier.
- Compared engagement levels to identify **most active subscribers**.

Insights

- **Premium users** usually show the **highest watch time** (higher engagement).
- Basic users tend to have **lower viewing activity**.
- Higher subscription tiers often indicate **more frequent usage**.

Analysis

- Helps Netflix understand **value of each subscription plan**.
- Supports strategies to **upgrade users from Basic to Premium**.
- Enables better **content and pricing decisions**.

Descriptive & Comparative Analysis

Top 10 Most Watched Shows/Movies

Aggregated **total watch time for each show/movie**.

Sorted data in **descending order** to identify top 10 content.

Compared overall watch time between **Movies and TV Shows**.

Insights

- Top 10 content contributes a **major share of total watch time**.
- **TV Shows** often have higher watch time due to multiple episodes.
- Movies may have **high short-term engagement**, but shows drive long-term viewing.

Analysis

- Helps Netflix invest in **high-performing content genres**.
- Supports decisions on **renewing popular shows**.
- Guides **content recommendation strategies**.



Genre Popularity by Country

Grouped data by **Country and Genre**.

Calculated **total viewers/watch count** for each genre in every country.

Selected the **top genre per country** based on highest viewership.

Insights

- Different countries show **different genre preferences**.
- Some genres (e.g., Drama, Action) are **globally popular**.
- Regional preferences reflect **cultural and audience behavior differences**.

Analysis

- Helps Netflix create **region-specific content strategies**.
- Supports **localized recommendations** for users.
- Guides investment in **country-specific productions**.



Customer Ratings Distribution

Grouped ratings into **1 to 5 star categories**.

Calculated **frequency of each rating level**.

Computed **average rating by Genre and Country**.

Insights

- Most ratings are usually **clustered around 3–5 stars**.
- Certain genres (e.g., Drama/Thriller) tend to have **higher average ratings**.
- Rating patterns vary across countries based on **audience preferences**.

Analysis

- Helps identify **high-quality and well-rated content**.
- Supports improving **low-rated genres or regions**.
- Enhances **recommendation systems and user satisfaction**.

Trending Content Analysis

- Filtered dataset to extract content added in the **last 2 years** using date_added.
- Segmented data into **New Content vs Older Content**.
- Calculated and compared **average ratings** for both groups.

Insights

- New content reflects **recent trends and audience preferences**.
- Older content may have **more stable and higher ratings** due to established popularity.
- Continuous addition of content keeps the platform **engaging and competitive**.

Analysis

- Helps Netflix focus on **recent trending content strategy**.
- Supports decisions on **content acquisition and production**.
- Balances **new releases vs classic content retention**.

Market Insights & Revenue Analysis

Subscription Market Share Analysis

Counted total users for each **subscription type**.

Calculated **percentage share** of each plan.

Compared subscription distribution **across different countries**.

Insights

- Identified the **most popular subscription plan** overall.
- Premium plans often generate **higher engagement and revenue**.
- Subscription preference varies by **country and economic factors**.

Analysis

- Helps optimize **pricing and subscription strategies**.
- Supports targeted **upgrade campaigns (Basic → Premium)**.
- Enables **country-specific subscription planning**.

Customer Retention Insights

- Counted number of shows watched by each customer.
- Categorized users into **Single-Show vs Multiple-Show viewers**.
- Compared **watch time with ratings** to identify engagement trends.

Insights

- Majority of retained users tend to **watch multiple shows**.
- Customers with higher watch time are **more engaged and loyal**.
- Heavy watchers often give **higher ratings**, indicating satisfaction.

Analysis

- Helps identify **loyal vs inactive users**.
- Supports strategies to **increase user retention**.
- Enables personalized recommendations to **boost engagement**.

High-Rated vs. Low-Rated Content

Calculated **average rating for each genre**.

Identified **top-rated and lowest-rated genres** based on average scores.

Compared **average ratings between TV Shows and Movies**.

Insights

- Certain genres (e.g., Drama/Documentary) often receive **higher ratings**.
- Some genres may consistently show **lower audience satisfaction**.
- **TV Shows** may have higher engagement, while **Movies** may have varied ratings.

Analysis

- Helps focus on **high-performing content genres**.
- Supports improving or reducing **low-performing genres**.
- Guides decisions on **content production and acquisition**.

Business Decision-Making Scenarios

Director & Cast Influence on Ratings

Grouped data by Director and Cast.

Calculated average rating for each director.

Compared ratings to identify impact of popular creators on content quality.

Insights

- Popular directors often produce **higher-rated content**.
- Strong cast presence can **positively influence audience ratings**.
- Top directors consistently deliver **high-quality content**.

Analysis

- Helps Netflix collaborate with **top-performing directors and actors**.
- Supports investment in **high-quality content production**.
- Enhances **content recommendation and promotion strategies**.

Content Localization Strategy

Analyzed **average ratings and watch time by country**.

Identified countries with **low engagement (low watch time) and low ratings**.

Compared performance across regions to detect **underperforming markets**.

Insights

- Some countries show **low watch time and low ratings**, indicating weak engagement.
- Content may not align with **local culture or preferences**.
- Lack of regional content can reduce **user interest and retention**.

Recommendations

Localized Content:Produce content in **local languages and** Focus on **regional culture & stories**

Genre Customization:Promote genres popular in that country and Reduce low-performing genres

Pricing Strategy:Adjust **subscription plans** based on region

Regional Marketing:Use **local influencers & promotions**

Content Partnerships:Collaborate with **local directors and actors**

Best Time to Release New Content

- Extracted **month from date_added** column.
- Counted number of titles added in each month.
- Calculated **average ratings for content released per month**.

Insights

- Identified months with **highest content releases** (peak upload periods).
- Compared ratings to find **months with highest audience satisfaction**.
- Some months may show **high releases but lower ratings** (quality vs quantity gap).

Analysis

- Helps determine **optimal release timing** for maximum engagement.
- Supports planning of **seasonal content strategies (festivals, holidays)**.
- Balances **content quantity and quality** for better performance.
"significantly improve content performance and audience engagement."

Customer Engagement Dashboard

Combined key metrics: **Watch Time, Ratings, Genre Popularity, Subscription Type.**

Designed visuals to represent **customer behavior and engagement patterns.**

Added **slicers** for dynamic filtering (Country, Age Group, Subscription Type).

Dashboard Components

Insights

- High watch time indicates **strong user engagement.**
- Popular genres vary across **regions and age groups.**
- Premium users often show **higher engagement levels.**

Analysis

- Helps track **customer engagement in real-time.**
- Enables **personalized recommendations and marketing.**
- Supports decision-making for **content and subscription strategies.**



Competitor Benchmarking Simulation

Analyzed **number of titles and average ratings by country**.

Identified regions with **low content availability or low ratings**.

Compared hypothetical competitor strategy focusing on **localized and high-rated content**.

Insights

- Some countries have **limited content availability**, affecting engagement.
- Lower ratings in certain regions indicate **content mismatch with audience preferences**.
- Competitors focusing on **local content and quality** may perform better.

Recommendations

- Increase **localized content production** in underperforming countries.
- Focus on **high-quality content to improve ratings**.
- Expand **regional partnerships with local creators**.
- Optimize **recommendation systems based on user preferences**.

Conclusion:

The analysis revealed key patterns in customer behavior, content performance, and subscription usage, enabling a clear understanding of Netflix's engagement trends.

Recommendations/Suggestions:

- **Improve Content Quality:** Focus on **high-rated genres** and Reduce low-performing content
- **Localized Content Strategy:** Create **region-specific content** and Use **local language & culture**
- **Increase Content in Low Markets:** Expand content library in **underperforming countries**
- **Personalization** Recommend content based on: Age group, Watch history
- **Subscription Optimization:** Offer **flexible pricing** in different countries
- Popular genres like Drama and Action drive more watch time and receive higher ratings.
- TV Shows generally generate more engagement compared to Movies.
- Netflix should focus on producing high-quality content in top-performing genres to maximize viewership.
- Introducing targeted content for different countries can improve engagement in underperforming regions.
- Releasing content during peak months and promoting Premium plans can further increase user retention and revenue.



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THANK YOU